



BITS Pilani

Pilani | Dubai | Goa | Hyderabad | Mumbai

**WORK INTEGRATED
LEARNING PROGRAMMES**

Introduction to DevOps

PPT Booklet

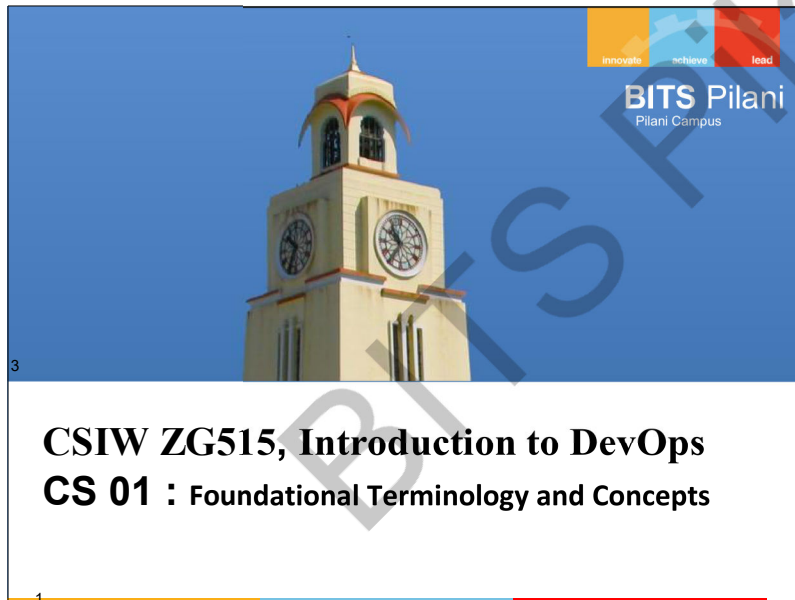


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**BITS Pilani
presentation**

Edited/Compiled by: Harvinder S Jabbal
WILP

1



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CSIW ZG515, Introduction to DevOps
CS 01 : Foundational Terminology and Concepts

1

IMP Note to Self



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IMP Note to Students

- ☐ It is important to know that just login to the session does not guarantee the attendance.
- ☐ Once you join the session, continue till the end to consider you as present in the class.
- ☐ IMPORTANTLY, you need to make the class more interactive by responding to Professors queries in the session.
- ☐ **Whenever Professor calls your number / name ,you need to respond, otherwise it will be considered as ABSENT**

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Coverage

- Software development lifecycle
- The Waterfall approach
- Agile Methodology
- Operational Methodologies: ITIL
- Development, Testing, Release, and Deployment Concepts
- Provisioning, Version Control
- Test Driven Development, Feature Driven Development
- Behavior-driven development

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Course Outline

Mod	Description
1	Why and What is DevOps?
2	DevOps Dimensions
3	Source Code Management (Using GIT as an example tool)
4	Continuous build and code quality
5	Continuous Integration and Continuous Delivery
6	Continuous Deployment
7	Continuous Monitoring
8	Configuration Management
9	Virtualization and Containerization

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Contact Session Wise Coverage

CS	Coverage
CS01	Process Models
CS02	Introduction to Devops
CS03	DevOps- Process/Agile Approach
CS04	DevOps – People/Technology/ Tools
CS05	Version Control Systems
CS06	Build Tools
CS07	Continuous Integration-Version control, automated build, Test
CS08	Artifacts management
CS09	Automating the deployment process
CS10	Rolling Back Deployments and Zero-Downtime Releases
CS11	Continuous monitoring
CS12	Managing Infrastructure and Environments
CS13	Virtualization and Containerization
CS14	Micro-services and Containerization
CS15	Current Trends- Kubernetes, DevOps on Cloud, Function-As-A-Service (AWS Lambda)
CS16	REVIEW

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Self Study Labs

S	Self Study Lab Exercise
1	Version Control using GIT, Creating repositories in GIT, Exercises to demonstrate the use of GIT operations and commands(Push, pull, clone etc.,) Creating branches and merging branches using GIT
2	Installation of Jenkins and Configuration of Jenkins to work with different version control, build and testing tools
3	Create jobs and projects in Jenkins
4	Continuous integration with Jenkins through source code polling and build triggers
5	Continuous inspection with Jenkins using sonarqube to ensure code quality
6	Continuous deployment/delivery to staging/production environment with Jenkins.

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Assignments

Skill Area	Activity	Activity
GIT	Install GIT	Create an account in github
Jenkins	Install Jenkins	Experiment with Jenkins as required in the Assignment.
Docket	Install Docker	Experiment with Docker as required in the Assignment.

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Evaluation of Assignment

You are expected to execute the Labs
You are expected to submit the Assignments.
Evaluation will be based on
Continuous Interaction
And Quizzes
During the Contact Sessions.
You must share you experience by requesting
Audio/Video and Presentation Access from faculty.
Faculty will not give opportunity to new participants
At the end of the course.

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Recorded Lectures for CS01

RL1.1 Foundational Terminology and Concepts

- [RL1.1.1 Agile Methodology](#)
- [RL1.1.2 Operational Methodologies: ITIL](#)

RL 1.2 Software Delivery

- [RL2.1.1 Problems of Delivering Software](#)
- [RL 2.1.2 Principles of Software Delivery](#)

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Recorded Lectures for CS02

RL 1.3 About DevOps

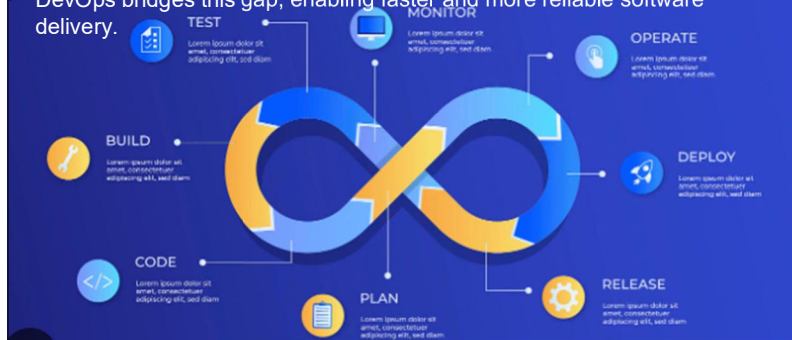
- [RL2.2.1 Need for DevOps](#)
- [RL2.2.2 Evolution of DevOps](#)
- [RL2.2.2 DevOps Practices](#)

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DevOps: Bridging the Gap Between Development and Operations

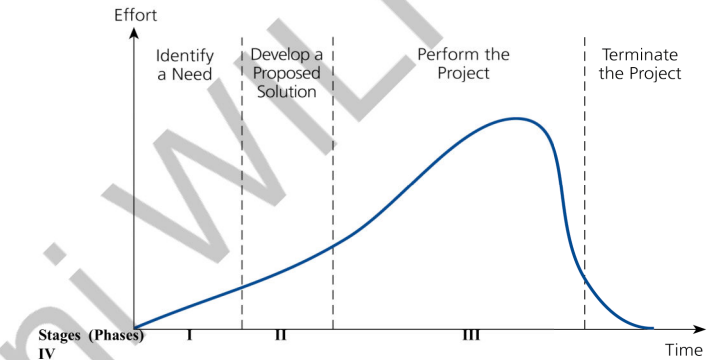
- Fosters collaboration between Development and Operations teams.
- Traditional siloed approach between Dev and Ops, leading to slow delivery and bottlenecks.
- DevOps bridges this gap, enabling faster and more reliable software delivery.



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Typical Project Life-Cycle (4 Phases)



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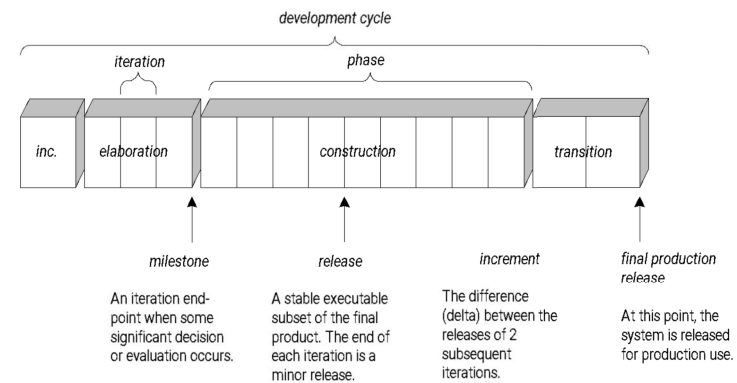
Software development lifecycle (Unified Process)

Inception
Elaboration
Construction
Transition

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UNIFIED PROCESS phases



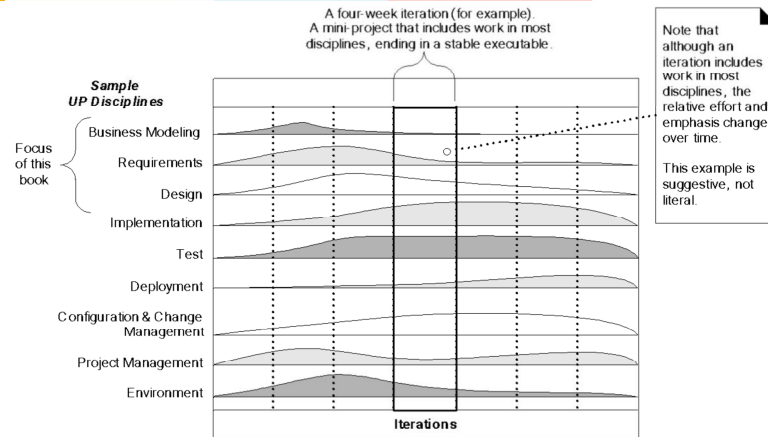
October 10, 2020

SEWI ZGS12 Object-Oriented Analysis and Design

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UNIFIED PROCESS disciplines



October 10, 2020

SEWI ZG512 Object Oriented Analysis and Design

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Waterfall Model Vs Iterative

Chunks

- Waterfall by activity
- Iterative by subsets of functionalities

Production Quality

Multiple Releases

Incremental, spiral, evolutionary, jacuzzi.

Compromise: Staged Delivery

Time Boxing

Slipping a date or slipping functionality.

October 17, 2020

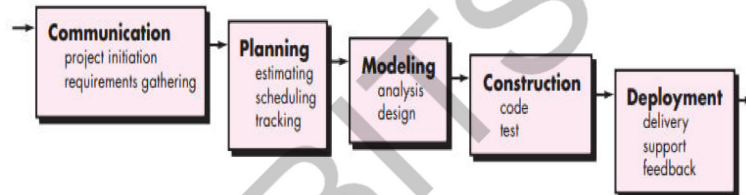
SEWI ZG512 Object Oriented Analysis and Design

18

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BITS Pilani, Deemed to be University under Section 3 of UGC Act, 1956

The Waterfall approach

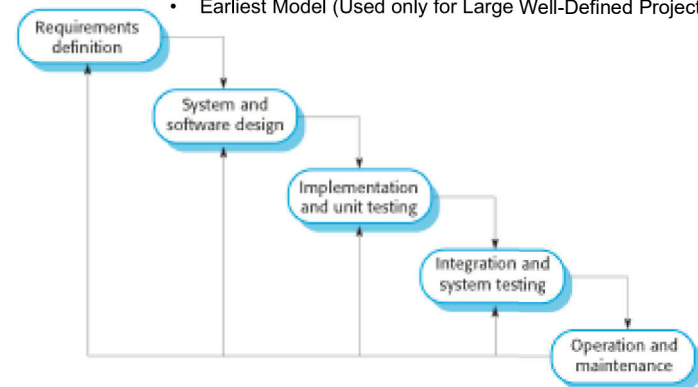


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Waterfall Model (Plan-Driven)

- Earliest Model (Used only for Large Well-Defined Projects)



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Agile Methodology

- Effective (rapid and adaptive) response to change
- Effective communication among all stakeholders
- Drawing the customer onto the team
- Organizing a team so that it is in control of the work performed
- *Yielding ...*
- Rapid, incremental delivery of software

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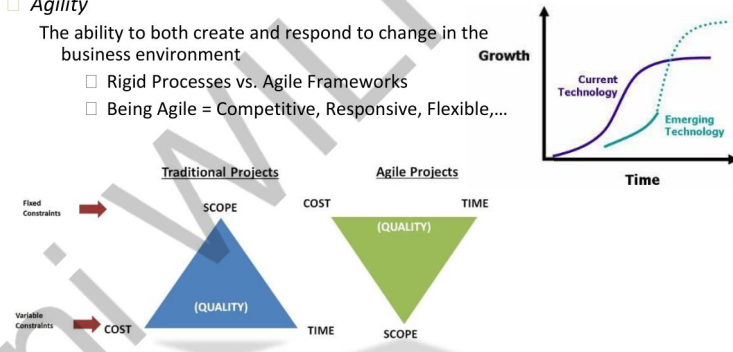
Agile Methods (Flexible, Adaptive)

- Adaptive to Changing or Evolving Customer Requirements

Agility

The ability to both create and respond to change in the business environment

- Rigid Processes vs. Agile Frameworks
- Being Agile = Competitive, Responsive, Flexible,...



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The Manifesto for Agile Software Development

"We are uncovering better ways of developing software by doing it and helping others do it. Through this work we have come to value:

- *Individuals and interactions over processes and tools*
- *Working software over comprehensive documentation*
- *Customer collaboration over contract negotiation*
- *Responding to change over following a plan*

That is, while there is value in the items on the right, we value the items on the left more."

Kent Beck et al

BSDCH ZC343 - Software Engineering

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Operational Methodologies:

ITSM (IT Service Management) vs. Service Delivery

SM: *Strategic* approach to design, deliver, manage and improve the way businesses use information technology (IT).

SM includes all the individual activities and processes that support a service throughout its lifecycle: from service management to change management, problem and incident management, asset management, and knowledge management.

ITSM and IT service delivery sometimes are used interchangeably

- **ITSM** emphasizes **IT service operation and improvement**
- IT service delivery focuses on **the quality of the work and meeting customer expectations.**
- **ITIL** (Information Technology Infrastructure Library) is a popular ITSM Framework (the terms ITSM and ITIL are used synonymously)

"The implementation and management of quality IT services that meet the needs of the business. IT service management is performed by IT service providers through an appropriate mix of people, process and information technology." - ITIL

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ITSM as a Business Enabler - ITIL

ITSM

Ensure the organization's business goals are supported by high-quality, cost-effective, value-adding IT services

Improve the quality of IT service provisioning

Reduce the long-term cost of IT service infrastructure

IT Infrastructure Library (ITIL) is a Framework described in the following volumes:

- Service Support
- Service Delivery
- Security Management
- Application Management
- ICT Infrastructure Management
- Planning to Implement Service Management
- The Business Perspective

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Core Principles of DevOps

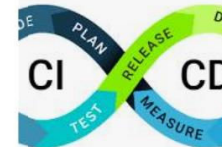
- Shared responsibility



- Automation



- Continuous integration and delivery (CI/CD)
- Infrastructure as code (IaC)
- Monitoring and feedback



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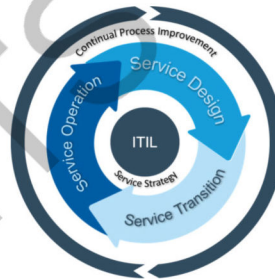
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Development, Testing, Release, and Deployment Concepts

Concept of Service Life Cycle

ITIL V3: Five Distinct Stages in ITSM

Service strategy
Service design
Service transition
Service operation
Continual service improvement.



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Provisioning, Version Control

Reproducibility.
Traceability.

These capabilities give teams several important benefits:

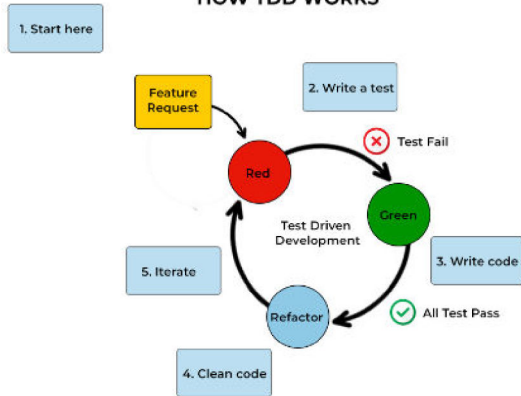
Disaster recovery.
Auditability.
Higher quality.
Capacity management.
Response to defects.

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Test Driven Development,

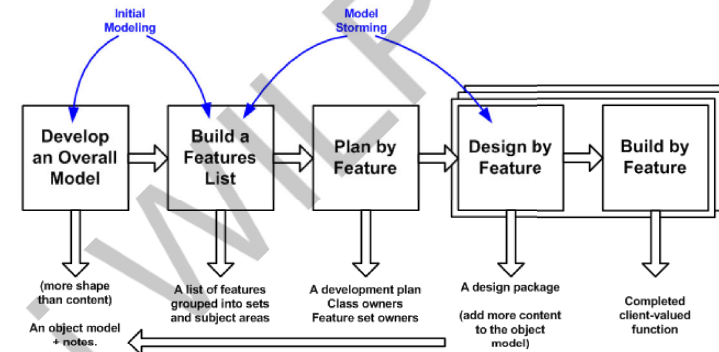
HOW TDD WORKS



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Feature Driven Development



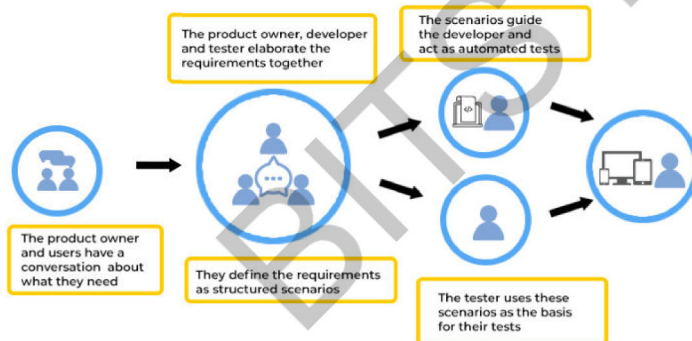
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Behavior-driven development

BDD DEVELOPMENT PROCESS

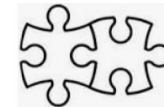


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Unleashing the Power of DevOps

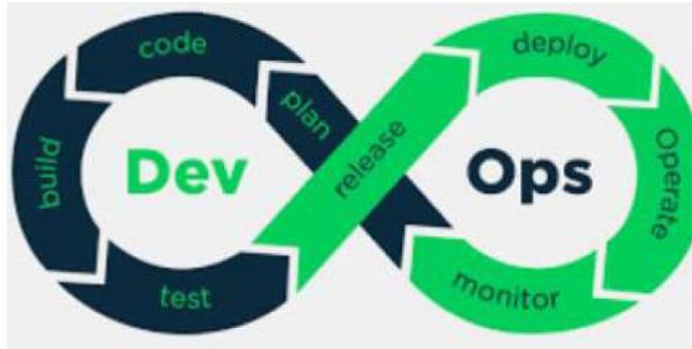
- Increased speed and agility.
- Improved quality and reliability
- Enhanced collaboration and communication
- Increased innovation



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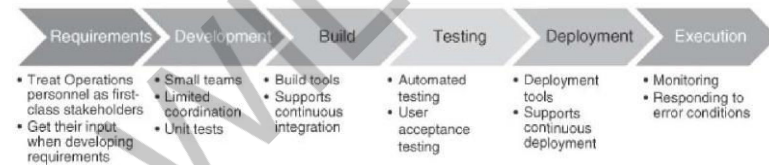
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The DevOps Life Cycle



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DevOps Toolbelt: Optimizing Your Workflow

- Version control systems (VCS)



- CI/CD tools



- Configuration management tools



- Monitoring tools



Prometheus

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Embrace the DevOps Journey

- Assess current state
- Define goals
- Build a DevOps culture
- Adopt DevOps tools
- Measure and iterate
- Call to action



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IMP Note to Self



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The End

Thank you

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Introduction to DevOps

Slides edited /compiled by Harvinder S Jabbal
for DevOps faculty team.

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Coverage

- Problems of Delivering Software
- Principles of Software Delivery
- Need for DevOps
- Evolution of DevOps- Define the stages of a DevOps evolution
- DevOps practices in organizations
- The Continuous DevOps LifeCycle Process (Continuous Integration, Continuous Inspection, Continuous Deployment, Continuous Delivery, Continuous Monitoring)
- DevOps Culture
- Case Study- (IBM/Facebook/Netflix)

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Recorded Lectures for CS03

RL2.1 Pillars of DevOps

[RL2.1.1 Three dimensions of DevOps – People, Process, Technology/Tools / DevOps Misconception](#)

[RL2.1.2 Agile Methodology – Scrum](#)

RL2.2 DevOps- Process

[RL2.2.1 DevOps and Agile](#)

[RL2.2.2 Agile methodology for DevOps Effectiveness](#)

[RL2.2.3 Behavior Driven Development, Feature Driven Development and Test Driven Development](#)

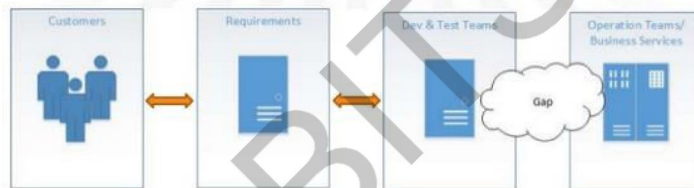
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Problems of Delivering Software

Why DevOps? – Delivery Challenges



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Delivery: Dev View

Why Gaps?

Dev View:

- Mostly delivers features after testing in development systems
- Dev systems may not be same as production system
- Developers will have faster turn around time w.r.t features
- Not much concerned about the infrastructural as well as deployment impact because of the code changes

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Delivery: Ops View

Why Gaps?

Ops View:

- Worries more about PSR
- Rewarded mainly for uptime
- Lesser turn around time w.r.t feature deployment and testing due to large number of dev builds coming their way
- Very much concerned about the infrastructural as well as deployment impact because of the code changes

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Dev and Ops dialogue

Dev:

- Put the current release live, NOW!
- It works on my machine
- We need this Yesterday
- You are using the wrong version



Ops:

- What are the dependencies?
- No machines available...
- Which DB?
- High Availability?
- Scalability?



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Problems of Delivering Software

Talking a different language

Formative experiences	Maturists (pre-1945)	Baby boomers (1945-1960)	Generation X (1961-1980)	Generation Y (1981-1995)	Generation Z (Born after 1995)
	War-time rationing Rock'n'roll Nuclear families Defined gender roles - particularly for women	Cold War 'Swinging Sixties' Moon landings Youth culture Woodstock Family-orientated	Fall of Berlin Wall Reagan/Gorbachev/Thatcherism Live Aid Early mobile technology Divorce rate rises	9/11 terrorists attacks Social media Invasion of Iraq Reality TV Google Earth	Economic downturn Global warming Mobile devices Cloud computing Wiki-leaks
Attitude toward career	Jobs for life	Organisational - careers are defined by employees	"Portfolio" careers - loyal to profession, not to employer	Digital entrepreneurs - work "with" organisations	Multitaskers - will move seamlessly between organisations and "pop-up" businesses
Signature product	Automobile	Television	Personal computer	Tablet/smartphone	Google glass, 3-D printing
Communication media	Formal letter	Telephone	E-mail and text message	Text or social media	Hand-held communication devices
Preference when making financial decisions	Face-to-face meetings	Face-to-face ideally but increasingly will go online	Online - would prefer face-to-face if time permitting	Face-to-face	Solutions will be digitally crowd-sourced

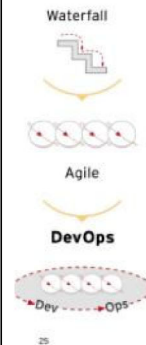
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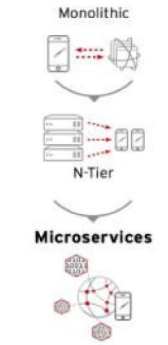
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Principles of Software Delivery

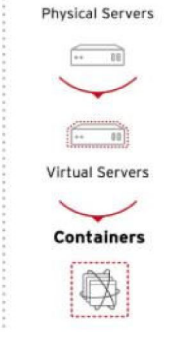
Development Process



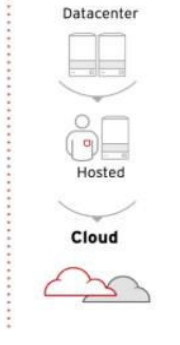
Application Architecture



Deployment & Packaging



Application Infrastructure



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Need for DevOps

What is DevOps?



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Evolution of DevOps- Define the stages of a DevOps evolution



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Evolution of DevOps- Define the stages of a DevOps evolution

DEPLOYMENTS AT AMAZON.COM

11.6s	1,079	10,000	30,000
Mean time between deployments (weekday)	Max number of deployments in a single hour	Mean number of hosts simultaneously receiving a deployment	Max number of hosts simultaneously receiving a deployment

Total number of files	1 billion
Number of source files	9 million
Lines of source code	2 billion
Depth of history	35 million commits
Size of content	86TB
Commits per workday	40,000



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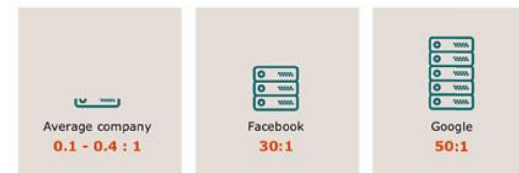
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Evolution of DevOps- Define the stages of a DevOps evolution

Operations: Scaling with bodies vs. software

Servers Per Employee (Timothy Chou, 2013)



How did they achieve this scale? By treating operations like a software problem.



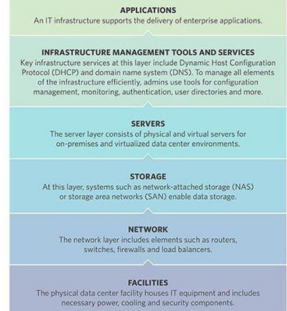
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LAYERS OF IT INFRASTRUCTURE



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Benefits of DevOps



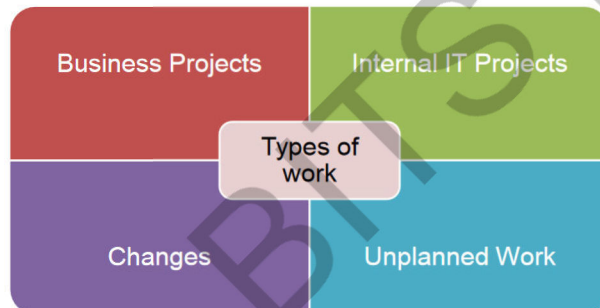
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DevOps practices in organizations

Systematically allocating time to the 4 types enables all the work to get done and becomes routine



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The Practice

- DevOps is a practice that:
 - Emphasizes the tight collaboration and communication of both software developers and IT operations staff
 - Focuses on automating the process of software delivery and infrastructure changes
 - Aims at establishing a culture and environment where building, testing, and releasing software, can happen rapidly, frequently, and more reliably

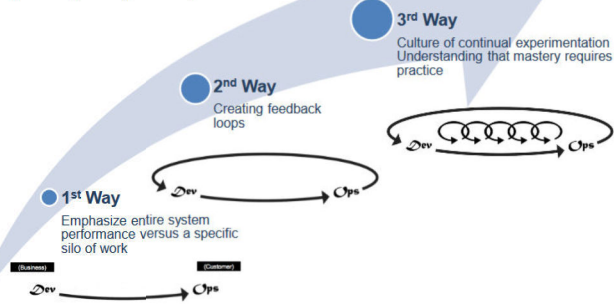
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How it works

The Three Ways describe the *values* that frame the *processes* of DevOps and they provide *prescriptive steps*



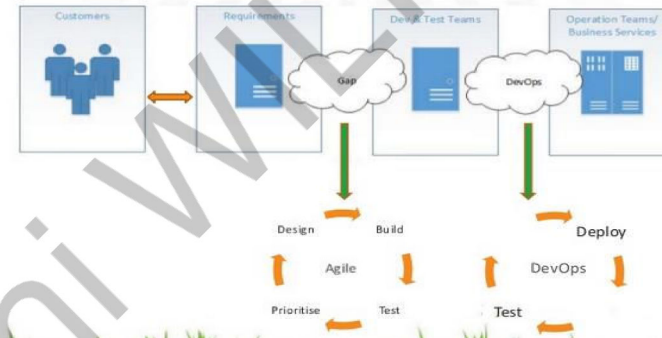
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Agile and DevOps

Agile and DevOps

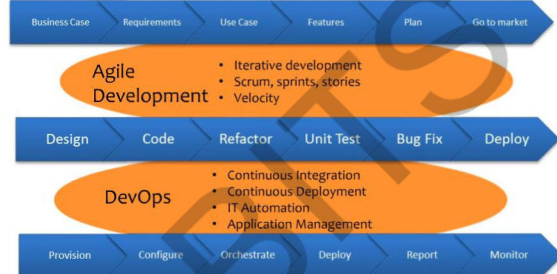


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Agile and DepOps



Business
Agility

IT
Agility

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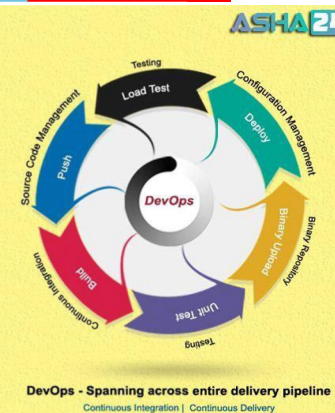
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The Continuous DevOps LifeCycle Process

DEVOPS LIFE CYCLE

- ✓ Push Code
- ✓ Fetch Changes
- ✓ Run Unit Tests
- ✓ Build Artifacts
- ✓ Store Artifacts
- ✓ Provision environment
- ✓ Deploy Your Build
- ✓ Run Load & Functional Tests
- ✓ Dev -> QA -> Staging -> Production



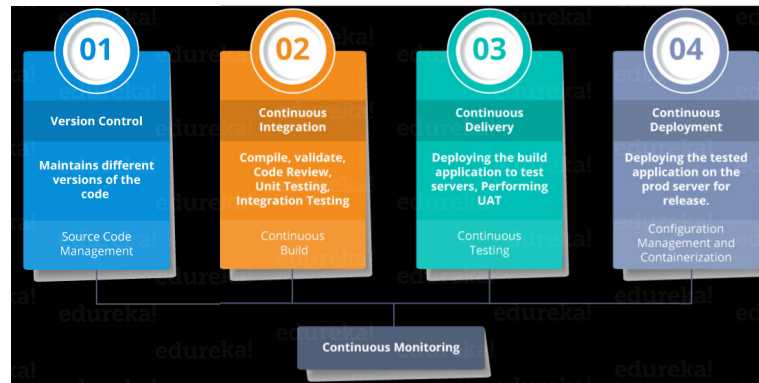
DevOps - Spanning across entire delivery pipeline
Continuous Integration | Continuous Delivery

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DevOps Lifecycle Processes

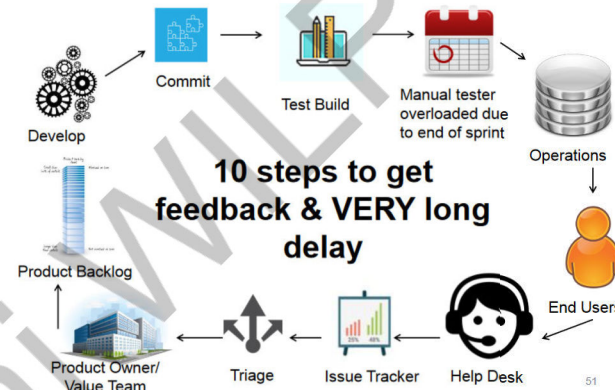


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Continuous Integration,

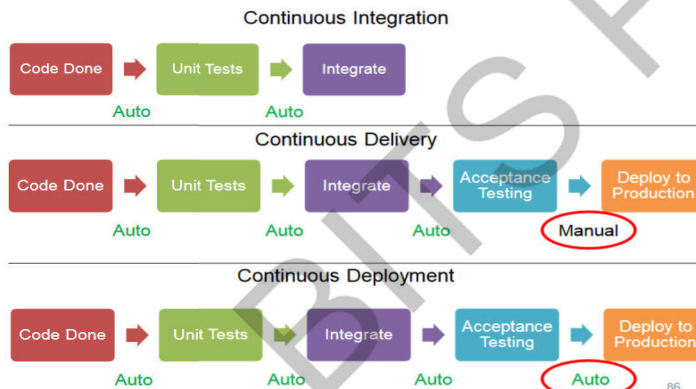


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Continuous Inspection,

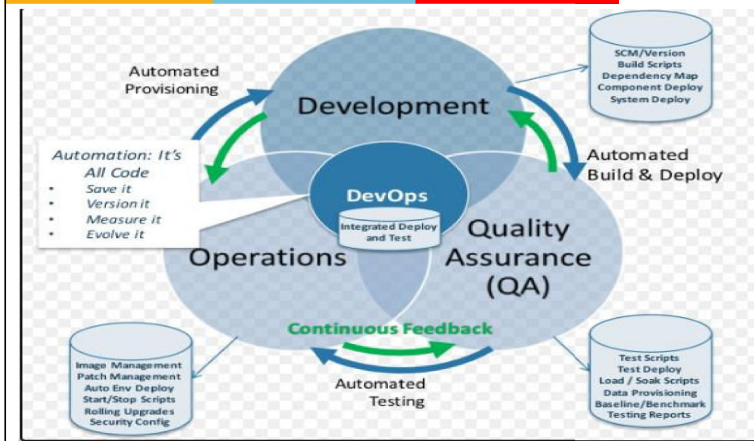


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Continuous Deployment,



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Continuous Delivery,

Deploy very frequently

**Practice,
Practice,
Practice**

Small deployments mean Fast deployments mean more deployments mean easier deployments mean lower cycle times means faster time to market

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Continuous Monitoring

What is a Feedback Loop?

A common DevOps term, feedback loops are a largely misunderstood concept Let's begin by addressing what the words "feedback" and "loop" mean, and then how they work together to complete useful work.

- Feedback is defined in the [Oxford Dictionary](#) as, "The modification or control of a process or system by its results or effects".
- Loop is defined as, "A structure, series, or process, the end of which is connected to the beginning."

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DevOps Culture

DELPHIX

Waterfall, Agile, Devops

- Waterfall



- Agile



- Agile with Continuous Deploy



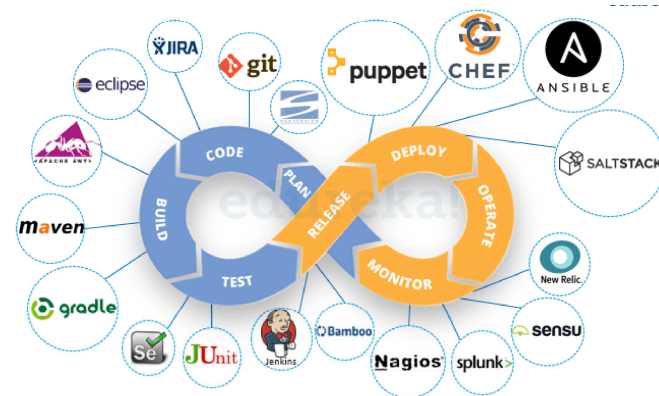
Continuous Deploy requires DevOps

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Continuous Integration, Continuous Inspection, Continuous Deployment, Continuous Delivery, Continuous Monitoring)



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DevOps Pipeline

A Pipeline is a chain of tasks that can be automated



- Integration tools use pipelines to perform tasks repetitively and continuously
- The process is called Continuous Integration (CI)
- Pipelines keep work flowing forward in our DevOps system

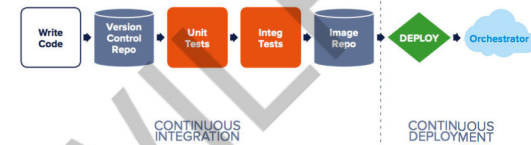
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Typical Case Study

A TYPICAL SOFTWARE DELIVERY PIPELINE

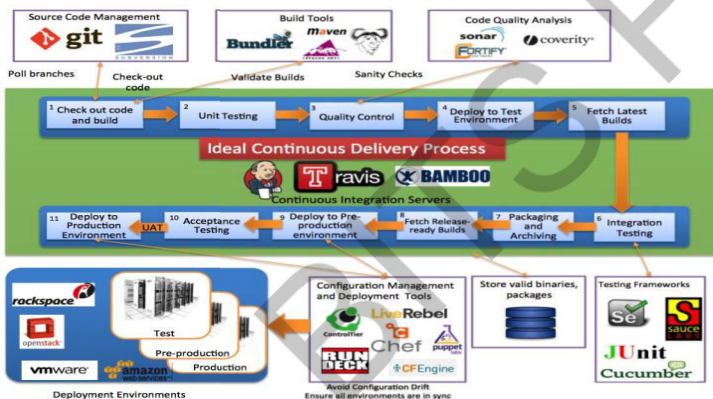


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Cse Study

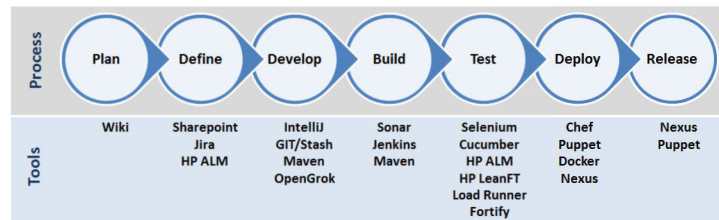


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Case Study



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Release Antipatterns

Common Quote “The day of a software release tends to be a tense one”

Antipattern: Deploying Software Manually

- The production of extensive and detailed documentation for steps to be performed
- Reliance on manual testing
- Frequent calls to the development team to explain
- Frequent corrections to the release process
- Sitting bleary-eyed in front of a monitor at 2 A.M

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Problems of Delivering Software Release Antipatterns Cond..

Antipattern: Deploying to a Production-like Environment Only after Development Is Complete

- Tester tested the system on development machines
- Releasing into staging is the first time that operations people interact with the new release
- Who Assembles? The Development Team
- Collaboration between development and Operations?

Antipattern: Manual Configuration Management of Production Environments

- Difference in Deployment to Stage and Production
- Different host behave differently
- Long time to prepare an environment
- Cannot step back to an earlier configuration of a system
- Modification to Configuration Directly

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Five different categories of DevOps practices

- Treat Ops as first-class citizens from the point of view of requirements
- Make Dev more responsible for relevant incident handling
- Enforce the deployment process used by all, including Dev and Ops personnel
- Use continuous deployment
- Develop infrastructure code, such as deployment scripts

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Principle of Continuous Delivery

Create a Repeatable, Reliable Process for Releasing Software

Automate Almost Everything

Keep Everything in Version Control

If It Hurts, Do It More Frequently, and Bring the Pain Forward

Build Quality In

“The Earlier you catch the defects, the cheaper they are to fix”

Done, Means Released

Everybody Is Responsible for the Delivery Process

Continuous Improvement

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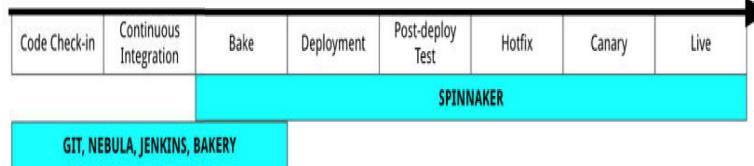
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NetFlix



<https://netflixtechblog.com/how-we-build-code-at-netflix-c5d9bd727f15>

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Facebook

Software development at Facebook runs contrary to many of the common practices of the industry. The main points we have covered include:

- There is no detailed plan to achieve a final, well-specified product.
- Engineers work directly on a common codebase with no branches and merging.
- There is no separate QA team responsible for testing.
- New code is released at a high rate, currently twice every working day.
- Engineers self-select what to work on.
- There is no assignment of blame for failures.

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facebook

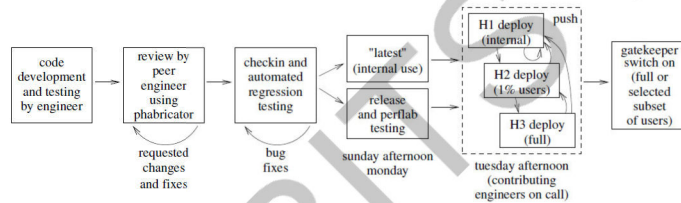


Figure 7: Facebook's version of the deployment pipeline, showing the multiple controls over new code.

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IMP Note to Self



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The End



Thank you

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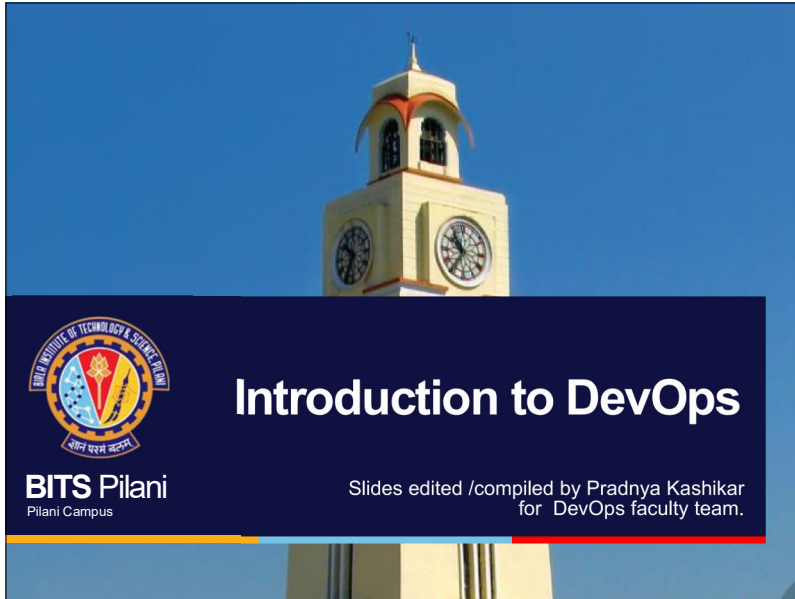
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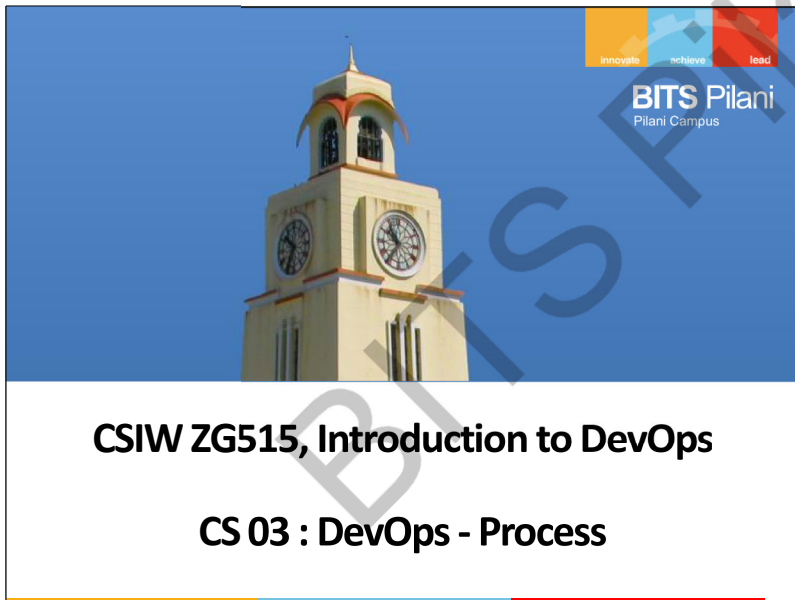
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**Introduction to DevOps**

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Slides edited /compiled by Pradnya Kashikar
for DevOps faculty team.



**CSIW ZG515, Introduction to DevOps**

CS 03 : DevOps - Process

IMP Note to Self



IMP Note to Students

- It is important to know that just login to the session does not guarantee the attendance.
- Once you join the session, continue till the end to consider you as present in the class.
- IMPORTANTLY, you need to make the class more interactive by responding to Professors queries in the session.
- **Whenever Professor calls your number / name ,you need to respond, otherwise it will be considered as ABSENT**



Coverage

• DevOps- Process

- Agile methodology for DevOps Effectiveness
- Flow Vs Non-Flow based Agile processes
- Choosing the appropriate team structure: Feature Vs Component teams
- Enterprise Agile frameworks and their relevance to DevOps
- Discuss (with examples and practical insights) Test Driven Development, Feature Driven Development, Behavior-driven development
- Cloud as a catalyst for DevOps

T1 - Chapter 2,

R1 - Chapter 4

Web technology for developers -
<https://developer.mozilla.org/en-US/docs/Web>

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Recorded Lectures for CS03

RL2.1 Pillars of DevOps

[RL2.1.1 Three dimensions of DevOps – People, Process, Technology/Tools](#)

[RL2.1.2 Agile Methodology - Scrum](#)

RL2.2 DevOps- Process

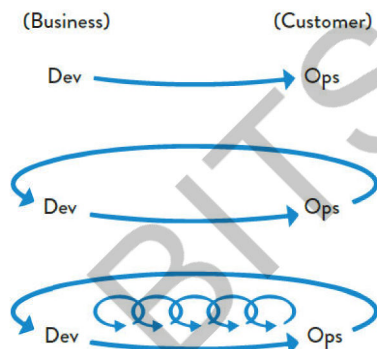
[RL2.2.1 DevOps and Agile](#)

[RL2.2.2 Agile methodology for DevOps Effectiveness](#)

[RL2.2.3 Behaviour Driven Development, Feature Driven Development and Test Driven Development](#)

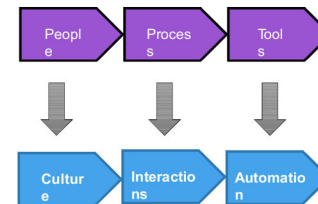
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“The Three Ways: The Principles Underpinning DevOps



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DevOps



• Process:

- Sequence or layers of procedures that is consisted of people, machines, materials, methods and governing policies that are used all together to design and delivery a good or service (product) . Conversion of an input into an output through-out procedures.

• Practice:

- Series of steps to achieve a consistent result (output).

- A simple process may be described by a single procedure. But a more complex process, will have multiple procedures.

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Dimensions of DevOps

□ Three dimensions of DevOps

□ People

- Initially, stakeholders would need to be identified, ensuring that everybody delivering value to the business are working tightly together on the common goal of adding value to the customer.

- Highly motivated people with good collaboration is necessary

□ Process

- Process of designing, building and testing software should be well presented to each individual team member
- Processes and practices that help improve the value and delivery to customers

□ Tools

Tools and services that can help enable different DevOps practices and different teams which can be used to make things easier

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Tools



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Tools



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Dispelling DevOps Myths

Corporate Culture

Development wants to add new features quickly

Operations want stability

Done! In two weeks the development is complete!!

You Wish!! We will take four weeks to get servers and deploy



The biggest business problem in the new fully digitalized world is not the cost of IT operations or the DevOps team – it is the lost opportunity on not executing your business service delivery with enough quality and speed compared with the new breed of competitors attacking your business segment.

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Dispelling DevOps Myths

Dev or Ops?

There is an issue with the build server.
Code isn't building.

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Dispelling DevOps Myths

Dev or Ops?

Software is deployed to the test environment.
You can't login to the application.

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Dispelling DevOps Myths

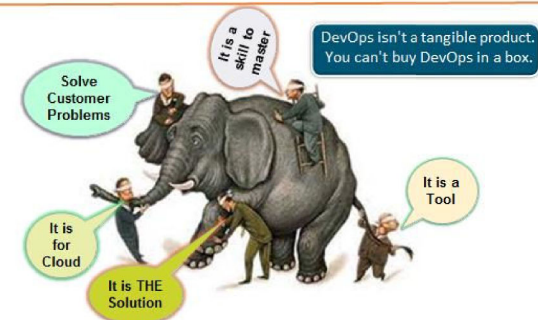


Dev vs. Ops

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Dispelling DevOps Myths

Popular Myths / Misconceptions



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Dispelling DevOps Myths

You Need a DevOps Guy...



DevOps is a methodology, not a skill. It is something you practice as a team not an individual.

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Dispelling DevOps Myths

DevOps supports any SDLC...



DevOps with legacy Waterfall or DevOps without Continuous Quality has a high risk of failure or not meeting customer satisfaction expectations.

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Dispelling DevOps Myths

DevOps Clashes with Existing Processes...



*DevOps smoothly integrates with existing processes like ITIL, Agile etc.
DevOps is actually a way to improve processes like ITIL.*

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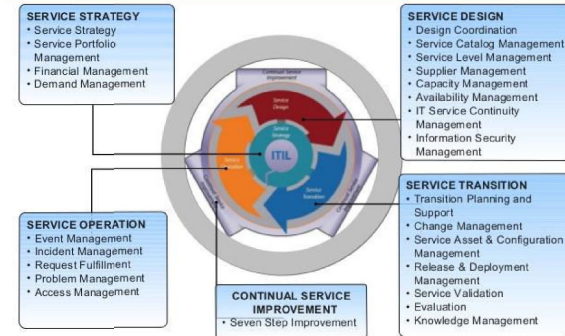
Food for Thought -

- What is the life cycle model followed in your project?
- What is your role name?
- What are the DevOps tools you use?
- Please share your experience

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ITIL

Lifecycle Processes

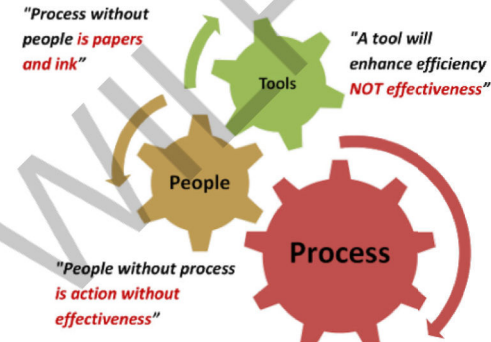


<Overview>

54

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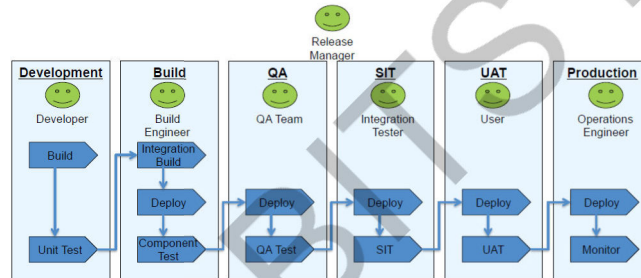
DevOps System



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A Typical Deployment Landscape

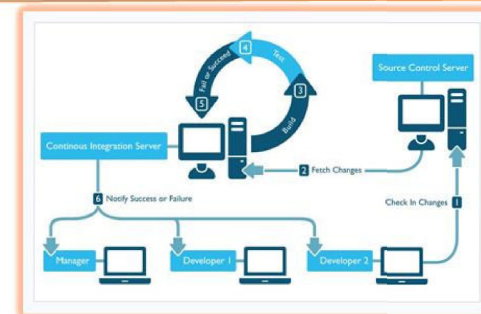
A Typical Deployment Landscape



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Test Driven Development

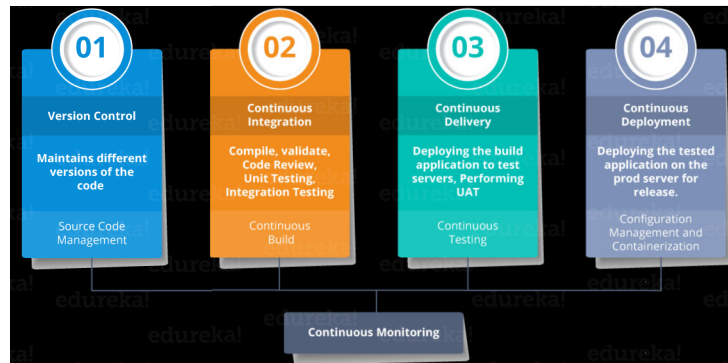
Software Development Life Cycle - Automation



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DevOps Life Cycle Process -

(Continuous Integration, Continuous Inspection, Continuous Deployment, Continuous Delivery, Continuous Monitoring)



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DevOps– Process

- Processes define what those people do
- organization can have a great culture of collaboration, but if people are **doing the wrong things** or **doing the right things in the wrong way**, failure is still likely.

KEY PROCESSES

(1) As a Business Process

- Makes the business more agile and improves its delivery of capabilities to customers.
- Business Process defines collection of activities or tasks that produces a specific result (service or product) for customers.
- DevOps business process involves taking capabilities from the idea (typically identified with business owners) through development and testing to production.
- Business process isn't mature enough to be captured in a set of simple process flows. Need to identify areas of improvement, both by improving the processes themselves and by introducing automation.

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DevOps– Process

(2) Change management process

- Set of activities designed to **control, manage, and track** change by identifying the work products that are likely to change and the processes used to implement that change.
- Traditional change management approaches tend to be limited to change request or defect management, with limited capability to trace the events between the change requests or defects and the associated code or requirements.
- Requires all stakeholders to be able to view and collaborate on all changes across the software development life cycle
- Organization uses is an inherent part of the broader DevOps process flow.
- Change management should include processes
 - Work-item management
 - Configurable work-item workflows
 - Project configuration management
 - Planning (agile and iterative)
 - Role-based artifact access control
- Includes processes that enable the enterprise to link work items to all artifacts, project assets, and other work items that are created, modified, referenced, or deleted by any practitioner who work on them.

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Agile methodologies for effective DevOps



Disciplined Agile Delivery phases for each release.

- Inception phase** - Release planning (Feature prioritization, scheduling) and initial requirements
- Construction phase** - management of the code branches, the use of continuous integration and continuous deployment
- ☐ **Transition phase** - solution is deployed and the development team is responsible for the deployment, monitoring the process of the deployment, deciding whether to roll back and when
- ☐ **Choosing appropriate Team Size** - size of the team should be relatively small
- ☐ **Team Roles** - Team lead (Scrum Master) Team member (Developer)

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